with an insistence that there is no absolute frame of motion leads us directly to the idea that time and space are "relative." Spacetime is universal and agreed upon by everyone, but how we divvy it up into "space" and "time" will be different for different observers.

Special relativity is a framework that includes many specific physical theories, all of which are dubbed "relativistic." Classical electromagnetism, put together by James Clerk Maxwell in the 1860s, is a relativistic theory even though it was invented before relativity; the need to better understand the symmetries of electromagnetism was a driving force behind why relativity was invented in the first place. (Sometimes people misuse the word "classical" to include "non-relativistic," but it's better to reserve it to mean "non-quantum.") Quantum mechanics and special relativity are 100 percent compatible with each other. The quantum field theories used in modern particle physics are relativistic to their cores.

The other big idea in relativity came ten years later, when Einstein proposed *general relativity*, his theory of gravity and curved spacetime. The crucial insight was that four-dimensional spacetime isn't just a static background on which the interesting parts of physics take place; it has a life of its own. Spacetime can bend and warp, and does so in response to the presence of matter and energy. We grow up learning about the flat geometry described by Euclid, in which initially parallel lines remain parallel forever and the angles inside a triangle always add up to 180 degrees. Spacetime, Einstein realized, has a non-Euclidean geometry, in which these venerable facts are no longer the case. Initially parallel rays of light, for example, can be focused together while moving through empty space. The effects of this warping of geometry are what we recognize as "gravity." General relativity came with numerous mind-stretching consequences, such as the expansion of the universe and the existence of black holes, though it has taken physicists a long time to appreciate what those consequences are.

Special relativity is a framework, but general relativity is a specific theory. Just like Newton's laws govern the evolution of a classical system or the Schrödinger equation governs the evolution of a quantum wave function, Einstein derived an equation that governs the curvature of spacetime. As with Schrödinger's equation, it's fun to actually see Einstein's equation written out, even if we don't bother with all the details:

$$R_{\mu\nu} - (\tfrac{1}{2})Rg_{\mu\nu} = 8\pi G T_{\mu\nu}$$